

UNIT 5 : Natural Language Processing

1. CHATBOT (Simple + Smart using Transformers)

📌 What this does

A chatbot that can respond like a human using a pretrained model.

```
# Install
!pip install transformers torch -q

from transformers import AutoModelForCausalLM, AutoTokenizer
import torch

# Load model
model_name = "microsoft/DialoGPT-medium"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)

# Fix: set pad token
tokenizer.pad_token = tokenizer.eos_token

print("Chatbot ready! Type 'exit' to stop.\n")

chat_history_ids = None

while True:
    user_input = input("You: ")

    if user_input.lower() == "exit":
        print("Chatbot ended.")
        break

    # Encode input
    new_input_ids = tokenizer.encode(user_input + tokenizer.eos_token, return_tensors='pt')

    # Create attention mask (IMPORTANT FIX)
    attention_mask = torch.ones(new_input_ids.shape, dtype=torch.long)

    # Append history
    if chat_history_ids is not None:
        input_ids = torch.cat([chat_history_ids, new_input_ids], dim=-1)
        attention_mask = torch.cat(
            [torch.ones(chat_history_ids.shape, dtype=torch.long), attention_mask],
            dim=-1
        )
    else:
        input_ids = new_input_ids

    # Generate response
    chat_history_ids = model.generate(
        input_ids,
        attention_mask=attention_mask,
        max_new_tokens=60,
        pad_token_id=tokenizer.eos_token_id,
        do_sample=True,
        top_k=50,
        top_p=0.95,
        temperature=0.7
    )

    # Decode response
    bot_output = tokenizer.decode(
        chat_history_ids[:, input_ids.shape[-1]:][0],
        skip_special_tokens=True
    )

    print("Bot:", bot_output)
```

Output:

```
... Loading weights: 100% 293/293 [00:00<00:00, 330.09it/s, Materializing param=transformer.wte.weight]
The tied weights mapping and config for this model specifies to tie transformer.wte.weight to lm_head.weight, but both are present in
GPT2LMHeadModel LOAD REPORT from: microsoft/DialogPT-medium
Key | Status |
-----+-----+
transformer.h.{0...23}.attn.bias | UNEXPECTED |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Chatbot ready! Type 'exit' to stop.

You: Hello
Bot: Hello!! :D
You: What is NLP
Bot: Network led programming
You: What is Natural Language Processing
Bot: It's a thing
You: Define Natural Language Processing
Bot: Something something Natural Language Processing
You: exit
Chatbot ended.
```

We use a **pre-trained chatbot model**

It remembers previous conversation using **chat_history**

It generates replies like a human

2. QUESTION ANSWERING (QA System)

📌 What this does

Given a paragraph + question → it finds the exact answer.

```
# Install
!pip install transformers -q

from transformers import pipeline

# Load QA model
qa_pipeline = pipeline("question-answering")

# Context (data)
context = """
Artificial Intelligence (AI) is a field of computer science that focuses on creating smart machines.
It is widely used in healthcare, finance, and education.
"""

# Question
question = "Where is AI used?"

# Get answer
result = qa_pipeline(question=question, context=context)

print("Question:", question)
print("Answer:", result['answer'])
```

Output:

No model was supplied, defaulted to `distilbert/distilbert-base-cased-distilled-squad` and revision `564e9b5`.
Using a pipeline without specifying a model name and revision in production is not recommended.

```
config.json: 100% 473/473 [00:00<00:00, 6.54kB/s]
```

model.safetensors: 100%  261M/261M [00:07<00:00, 41.4MB/s]

Loading weights: 100% 102/102 [00:00<00:00, 230.41it/s, Materializing param=qa_outputs.weight]

tokenizer_config.json: 100% 49.0/49.0 [00:00<00:00, 2.37kB/s]

vocab.txt: 213k/? [00:00<00:00, 2.67MB/s]

tokenizer.json: 436k/? [00:00<00:00, 7.39MB/s]

Question: Where is AI used?

Answer: healthcare, finance, and education

Explanation

- Model reads **context**
- Then finds the **exact answer span**
- No need to train anything

3. TEXT SUMMARIZATION



 **What this does**

Converts long text → short summary

```
[1] 25s
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load model and tokenizer (BART - best for summarization)
model_name = "facebook/bart-large-cnn"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

print("Model loaded successfully!\n")

# Input text
text = """
Artificial Intelligence is transforming the world by enabling machines to perform tasks that normally require human intelligence.
It is widely used in healthcare, finance, education, and robotics.
AI helps in automation, decision-making, and improving efficiency across industries.
"""

# Tokenize input
inputs = tokenizer(text, return_tensors="pt", max_length=1024, truncation=True)

# Generate summary
summary_ids = model.generate(
    inputs["input_ids"],
    max_length=60,
    min_length=20,
    length_penalty=2.0,
    num_beams=4,
    early_stopping=True
)

# Decode summary
summary = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Original Text:\n", text)
print("\nSummary:\n", summary)
```

Output:

```
... The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operation. You can interrupt this and resume the migration later on by calling 'tran
0.0 [00:00<7.7ms]
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: 'resume_download' is deprecated and will be removed in version 1.0.0. Downloads always resume when possible.
warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
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warnings.warn(
generation_config.json: 100% 363/363 [00:00<00:00, 25.9kB/s]
Model loaded successfully!

Original Text:
Artificial Intelligence is transforming the world by enabling machines to perform tasks that normally require human intelligence.
It is widely used in healthcare, finance, education, and robotics.
AI helps in automation, decision-making, and improving efficiency across industries.

Summary:
Artificial Intelligence is transforming the world by enabling machines to perform tasks that normally require human intelligence. It is widely used in healthcare, finance, education and robotics.
```

Explanation

- Reduces text size
- Keeps only important points
- Uses pretrained model

4. MACHINE TRANSLATION

 What this does

Converts one language → another language

```
[2]
✓ 33s # Install
!pip install transformers sentencepiece -q

from transformers import pipeline

# Load translator (English → French)
translator = pipeline("translation_en_to_fr")

# Input text
text = "Artificial Intelligence is changing the world."

# Translate
translation = translator(text)

print(["Original:", text])
print("Translated:", translation[0]['translation_text'])
```

Output:

```
... No model was supplied, defaulted to google-t5/t5-base and revision 686f1db (https://huggingface.co/google-t5/t5-base).
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated and
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spiece.model: 100% 792k/792k [00:00<00:00, 12.0MB/s]
tokenizer.json: 1.39M/? [00:00<00:00, 30.7MB/s]
Original: Artificial Intelligence is changing the world.
Translated: L'intelligence artificielle change le monde.
```

Explanation

- Uses trained translation model
- Converts sentence automatically
- No manual grammar rules needed

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/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>)
You will be able to reuse this secret in all of your notebooks.
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Original Text:

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Summary:

Artificial Intelligence is transforming the world by enabling machines to perform tasks that normally require human intelligence.

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Original: Artificial Intelligence is changing the world.
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Start coding or [generate](#) with AI.